ABSOLUTE FLATNESS AND REDUCEDNESS TRANSFER ALONG A RING MAP WHOSE TENSOR SQUARE IS FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

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Editorial note. The counted proof span opens with the author's sentence 'Part (1) follows immediately from Lemma ... and the definitions.'; that sentence is delivered verbatim and part (1) is not expanded. The five references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The key statement relating absolute flatness of A and B to flatness of B tensor_A B -> B. invoked at proof line 30332. Gives part (1) outright and is applied to A' -> B' in part (2).
- A reduced ring embeds into the product of its residue fields at minimal primes, and a product of fields is absolutely flat. invoked at proof lines 30334-30337. Produces the absolutely flat ring A' receiving A.
- Base change of a weakly etale ring map, and the statement that an absolutely flat ring is reduced. invoked at proof lines 30339-30342. Shows A' -> B' is weakly etale and hence B' is reduced.

Source reference. Original lemma 00EW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 550–550. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 30250: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30250–30250. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 30303: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30303–30303. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 30202: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30202–30202. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 30289: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30289–30289. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $A \to B$ be a ring map such that $B \otimes_A B \to B$ is flat.*

(1) *If A is an absolutely flat ring, then so is B.*
(2) *If A is reduced and $A \to B$ is weakly étale, then B is reduced.*

Proof. Part (1) follows immediately from Lemma 30202 and the definitions. If A is reduced, then there exists an injection $A \to A' = \prod_{\mathfrak{p} \subset A \text{ minimal}} A_{\mathfrak{p}}$ of A into an absolutely flat ring (Algebra, Lemma 00EW and Lemma 30289). If $A \to B$ is flat, then the induced map $B \to B' = B \otimes_A A'$ is injective too. By Lemma 30303 the ring map $A' \to B'$ is weakly étale. By part (1) we see that B' is absolutely flat. By Lemma 30250 the ring B' is reduced. Hence B is reduced. □